EXPRESS-G example model Car Registration Authority

Introduction

This document contains a complete EXPRESS-G representation of the Car Registration Authority example model. There are details which cannot be expressed using the EXPRESS-G iconic language but which can be described using the EXPRESS lexical language.

An EXPRESS model is a primary representation and an EXPRESS-G model is a secondary representation.

Model overview

The model consists of three schemas, as shown in Figure Complete schema-level model for Registration Authority example (Page 1 of 1)..

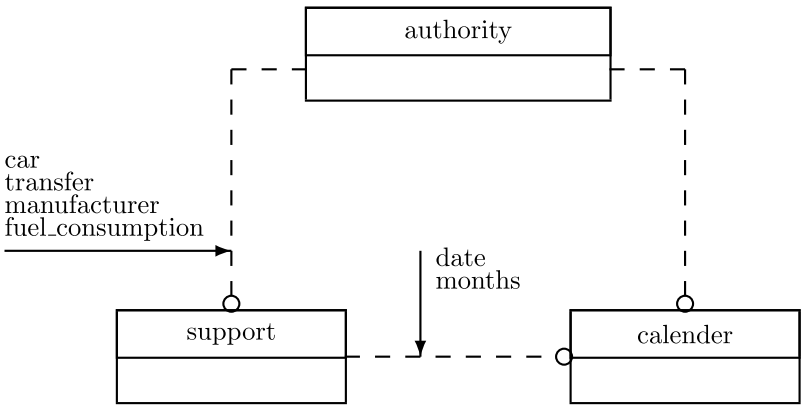


Figure 1. Complete schema-level model for Registration Authority example (Page 1 of 1).

The schema `tt "registration_authority"` is the primary schema. It references the noted items from the `tt "auxiliary"` schema; some functions, rules or procedures may also be referenced from this schema as well, but the diagram cannot show these. This schema also references one or more items from the `tt "calendar"` schema. These may be rules, functions or procedures but again the diagram cannot show these; alternatively, the entire schema may be referenced.

The `tt "auxiliary"` schema references at least two items from the `tt "calendar"` schema. Rules, functions and procedures may also be referenced, but this cannot be discerned via EXPRESS-G.

The `tt "calendar"` schema is self-contained as it neither uses nor references any other schema.

An entity-level model for the `tt "registration_authority"` schema is given in Figure Complete entity-level model of the Registration Authority schema (Page 1 of 1).. The entity-level model for the `tt "auxiliary"` schema is displayed in Figures Complete entity-level model of the Auxiliary schema (Page 1 of 2). and Complete entity-level model of the Auxiliary schema (Page 2 of 2).. The internals of the `tt "calendar"` schema are shown in Figure Complete entity-level model of Calendar schema (Page 1 of 1)..

The registration authority schema

Figure Complete entity-level model of the Registration Authority schema (Page 1 of 1). shows the `tt "registration_authority"` schema and the entities and types in it.

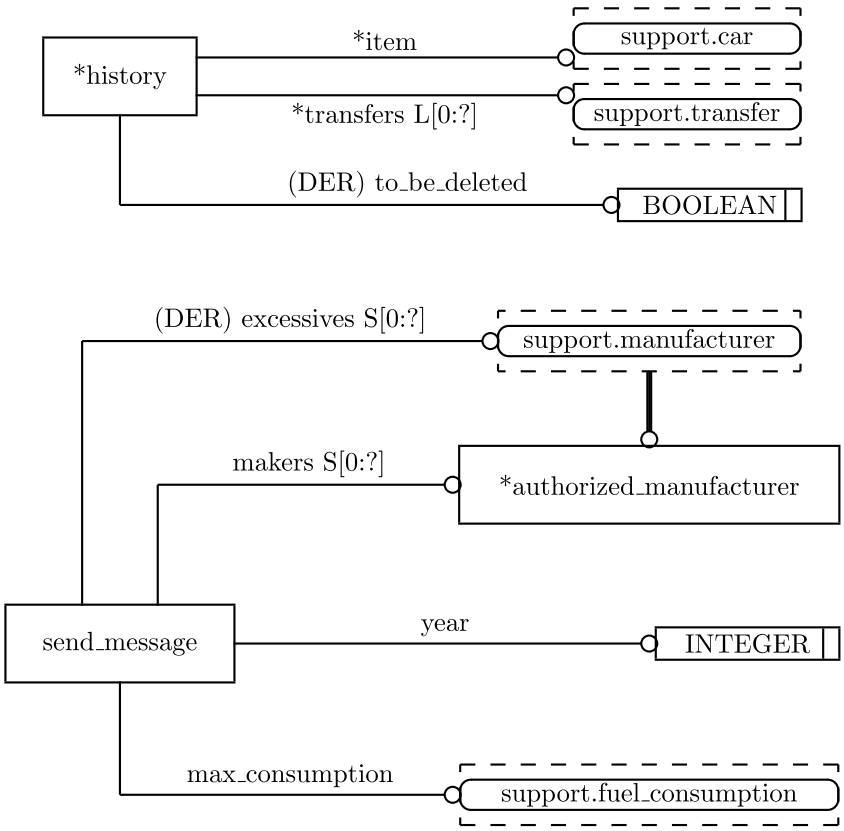


Figure 2. Complete entity-level model of the Registration Authority schema (Page 1 of 1).

There are three entities altogether, namely `tt "send_message"`, `tt "history"` and `tt "authorized_manufacturer"`. There are constraints on the two latter entities. The entity `tt "authorized_manufacturer"` is a subtype of the entity `tt "manufacturer"` which is defined in the `tt "auxiliary"` schema.

There are no types defined within the schema.

The auxiliary schema

The `tt "auxiliary"` schema contains too many items to be displayed on a single page. Figures Complete entity-level model of the Auxiliary schema (Page 1 of 2). and Complete entity-level model of the Auxiliary schema (Page 2 of 2). show the entities and types that are contained in the schema.

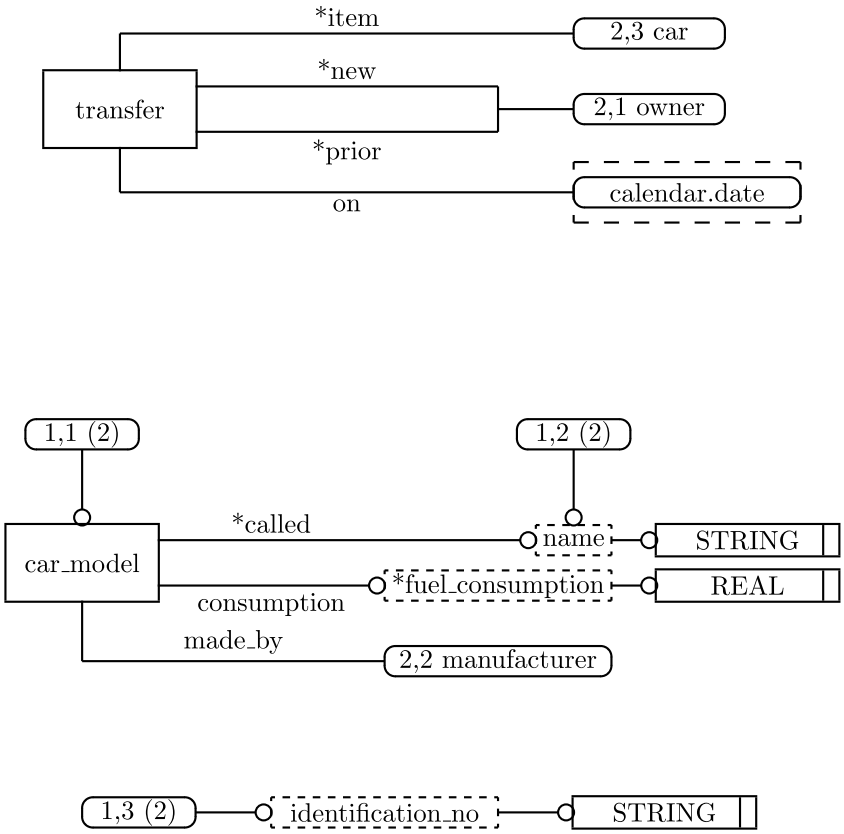


Figure 3. Complete entity-level model of the Auxiliary schema (Page 1 of 2).

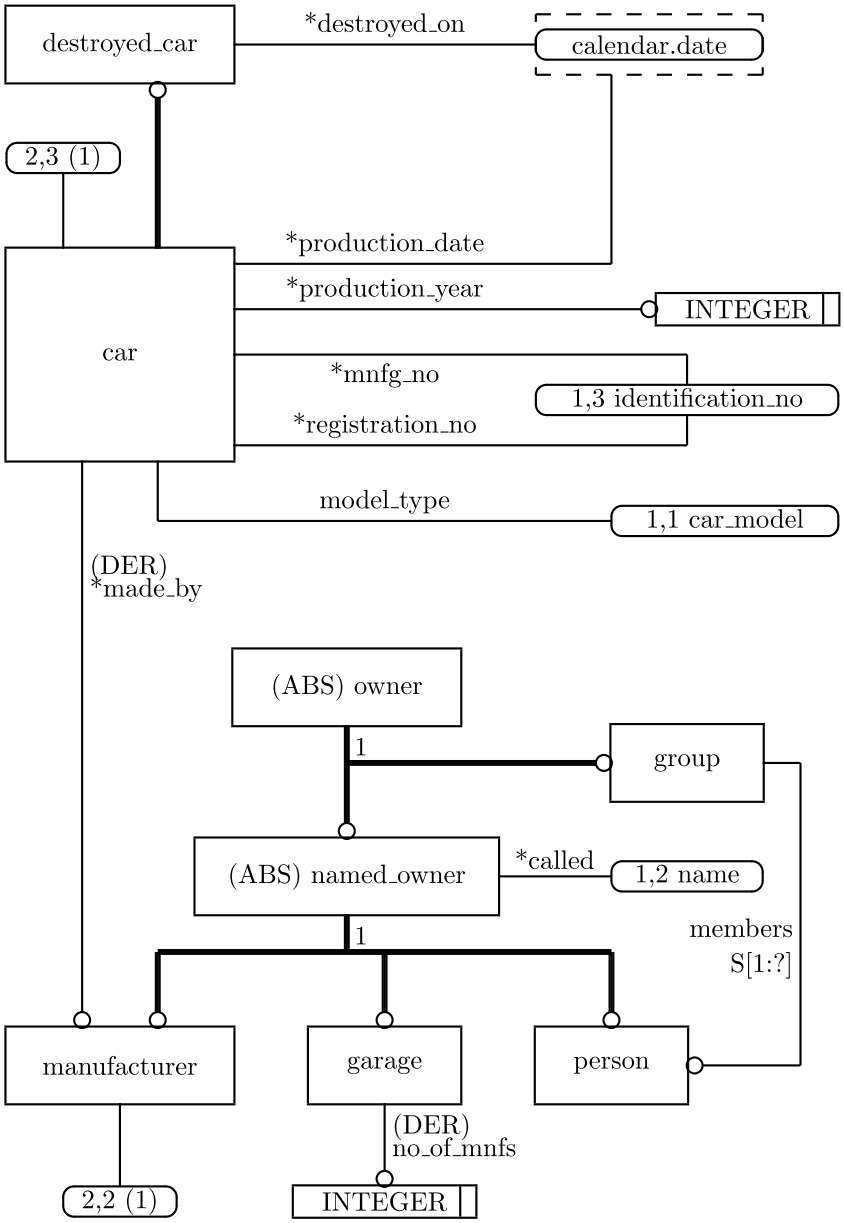


Figure 4. Complete entity-level model of the Auxiliary schema (Page 2 of 2).

The schema contains the following entities:

- `tt "car"`
- `tt "car_model"`
- `tt "destroyed_car"`, which is a subtype of `tt "car"`
- `tt "garage"`, which is a subtype of `tt "named_owner"`
- `tt "group"`, which is a subtype of `tt "owner"`
- `tt "manufacturer"`, which is a subtype of `tt "named_owner"`
- `tt "named_owner"`, which is an abstract supertype and also a subtype of `tt "owner"`. It puts a `tt "OneOf"` constraint on its subtypes.
- `tt "owner"`, which is an abstract supertype and puts a `tt "OneOf"` constraint on its subtypes.
- `tt "person"`, which is a subtype of `tt "named_owner"`
- `tt "transfer"`

It also contains the following types:

- `tt "identification_no"`, a defined data type with a base-type of `tt "String"`
- `tt "fuel_consumption"`, a defined data type with a base-type of `tt "Real"`
- `tt "name"`, a defined data type with a base-type of `tt "String"`

The calendar schema

Figure Complete entity-level model of Calendar schema (Page 1 of 1). shows the entities and types within the `tt "calendar"` schema.

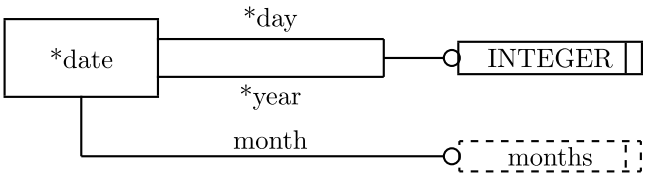


Figure 5. Complete entity-level model of Calendar schema (Page 1 of 1).

There is one entity called `tt "date"` and one enumeration type called `tt "months"`.